Brain Tumor Classification using Self-Supervised Contrastive Learning (SimCLR)

MIC Course Project

Medical Image Computing

Outline

Why is this hard?

The clinical problem

- Brain tumors require early, accurate diagnosis
- MRI is the gold standard imaging tool
- Radiologists manually inspect every slice — slow, expensive, fatigue-prone

The deep learning bottleneck

- Supervised DL needs lots of labelled data
- Medical labelling = expert radiologist time
- Annotation is the scarcest resource

Our Question

Can we learn good representations
without any labels
and still classify tumors well?

Our Answer

Yes — using SimCLR
94.13% test accuracy
with zero labelled pre-training

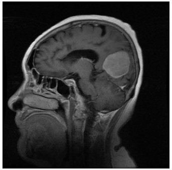
Brain Tumor MRI Dataset

Key facts

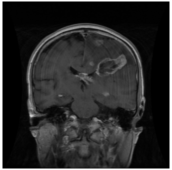
- **3,064** T1-weighted contrast-enhanced MRI slices
- **233 patients**, 3 tumor types
- Collected by Jun Cheng, Southern Medical University
- Stored as `.mat` files (MATLAB format)

Class	Slices	%
Meningioma	708	23.1%
Glioma	1426	46.6%
Pituitary	930	30.4%
Total	**3064**	100%

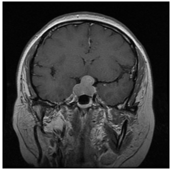
Sample MRI slices from dataset



Meningioma
Large posterior mass



Glioma
Ring-enhancing lesion



Pituitary
Small mass at skull base

Bright enhanced areas = tumor

T1-w contrast-enhanced MRI; axial/coronal/sagittal views

Patient-Level Data Split

The data leakage problem

- One patient has **multiple slices**
- A random image-level split lets the same patient appear in both train & test
- Model memorizes the patient, not the tumor
- Accuracy looks great — but it's **fake**

Our fix: stratified patient split

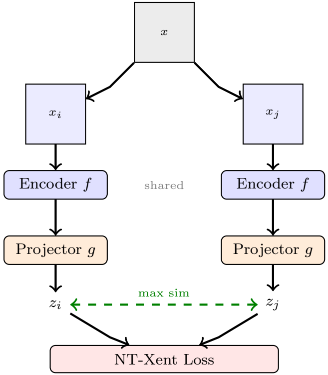
1. Group all 233 patients by class
2. Shuffle and split *patients*: 70% train / 15% val / 15% test
3. All slices of a patient go to *one* split only

Model sees **new patients** at test time — the realistic clinical scenario

What is SimCLR?

Core intuition

Two augmented views of the same image should be mapped ***close together****.*
Views from different images should be ***far apart****.*



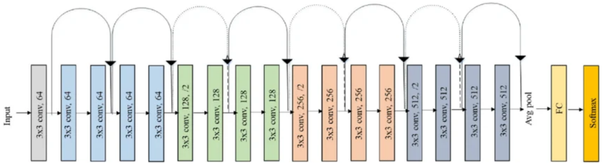
Key properties

- No labels needed during pre-training
- Self-supervised — labels are implicit (positive = same image, negative = everything else)
- Scales with batch size and data
- Chen et al., ICML 2020

The Encoder: ResNet-18

Why ResNet-18?

- Strong baseline backbone, well understood
- 18 layers with **residual (skip) connections**
- Skip connections let gradients flow directly — avoids vanishing gradient
- Outputs a **512-dim feature vector** after global average pooling
- We replace the final classification head with `nn.Identity()` and attach our projector



4 residual blocks: 64→128→256→512 channels. Dashed arrows = strided convolutions (downsampling). Solid curved arrows = identity skip connections.

SimCLR: Step by Step

tep 1 **Augment** — take image x, apply two random transforms t_1, t_2 to get views x_i, x_j

- Random crop, horizontal flip, normalize

tep 2 **Encode** — pass both views through shared ResNet-18

$$h_i = f(x_i) \in \mathbb{R}^{512}$$

tep 3 **Project** — 2-layer MLP maps to 128-dim unit sphere

$$z_i = \text{normalize}(g(h_i)) \in \mathbb{R}^{128}$$

tep 4 **NT-Xent Loss** — pull positives together, push negatives apart

$$\ell(i,j) = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}$$

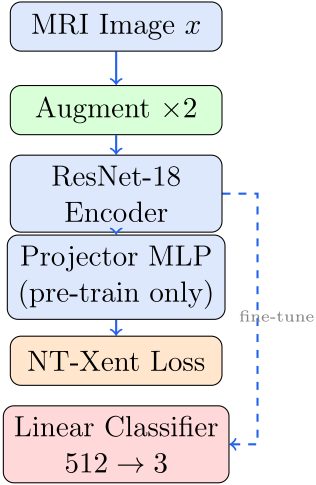
From Pre-training to Classification

Pre-training (no labels)

- 50 epochs, batch size 128
- ResNet-18 + projection head
- NT-Xent loss, $\tau = 0.5$
- Adam, lr $= 10^{-3}$
- Mixed precision (AMP)

Fine-tuning (with labels)

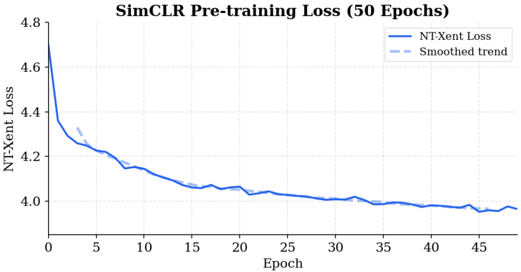
- **Discard** the projector head
- Attach linear layer: $512 \rightarrow 3$
- Fine-tune entire network end-to-end
- 20 epochs, lr $= 10^{-4}$



Pre-training Loss

What we see

- Loss starts at ≈ 4.53 (random features)
- Drops steadily to 3.86 by epoch 49
- Some noise mid-training — normal for contrastive learning
- Convergence confirms the encoder is learning meaningful structure



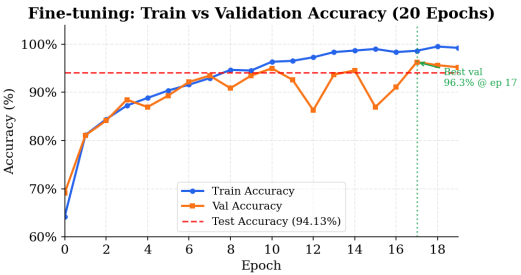
Interpretation

The encoder is learning to separate different MRI scans in feature space, *without ever seeing a label.*

Fine-tuning Accuracy

What we see

- Jumps from 64% to 93% in the first 7 epochs
- Train saturates near 99% by epoch 19
- Val peaks at 96.30%, more volatile — some overfitting
- Best val: **96.30%** at epoch 17



Test Accuracy

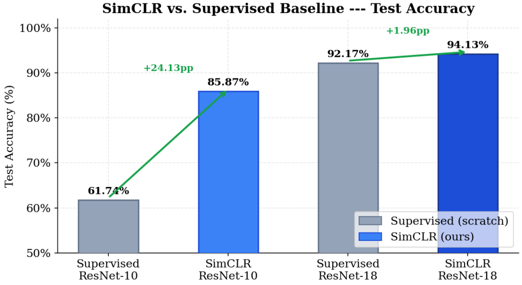
94.13%

on fully held-out patients

How Good is 94.13%?

What makes this impressive

- ResNet-18 trained **from scratch** (no ImageNet weights)
- Pre-training used **zero labels**
- Only 20 fine-tuning epochs
- Strict patient-level split — no leakage



Method	Backbone	Test Acc.
Supervised	ResNet-10	61.74%
SimCLR	ResNet-10	85.87%
Supervised	ResNet-18	92.17%
SimCLR	**ResNet-18**	**94.13%**

What Worked and What Could Be Better

What worked well

- ✓ SimCLR pre-training converged cleanly
- ✓ Patient-level split — fair evaluation
- ✓ Strong test accuracy from scratch
- ✓ Mixed precision made training fast

Possible improvements

- Stronger augmentations (Gaussian blur, jitter on repeated channels)
- Larger backbone (ResNet-50)
- Longer pre-training
- Early stopping to fix train/val gap
- Class-balanced sampling for meningioma
- Try MoCo or BYOL for comparison

The val/test gap

Val accuracy peaked at 96.30% but test hit 94.13%. This is likely due to random variation in the patient split — the test set may have ended up with slightly more typical cases.

Conclusion

Key takeaways

1. Self-supervised learning works well for medical imaging
2. SimCLR learns meaningful MRI features with **no labels**
3. Fine-tuning those features gives **94.13% accuracy** on 3-class tumor classification
4. Patient-level splitting is critical for honest evaluation

Broader impact

SSL can dramatically reduce the annotation burden in clinical settings, enabling models to leverage large unlabelled scan archives.

Resources

Dataset: Jun Cheng, Southern Medical University

Paper: Chen et al., "A Simple Framework for Contrastive Learning" ICML 2020 `arxiv.org/abs/2002.05709`

Code: Available on request